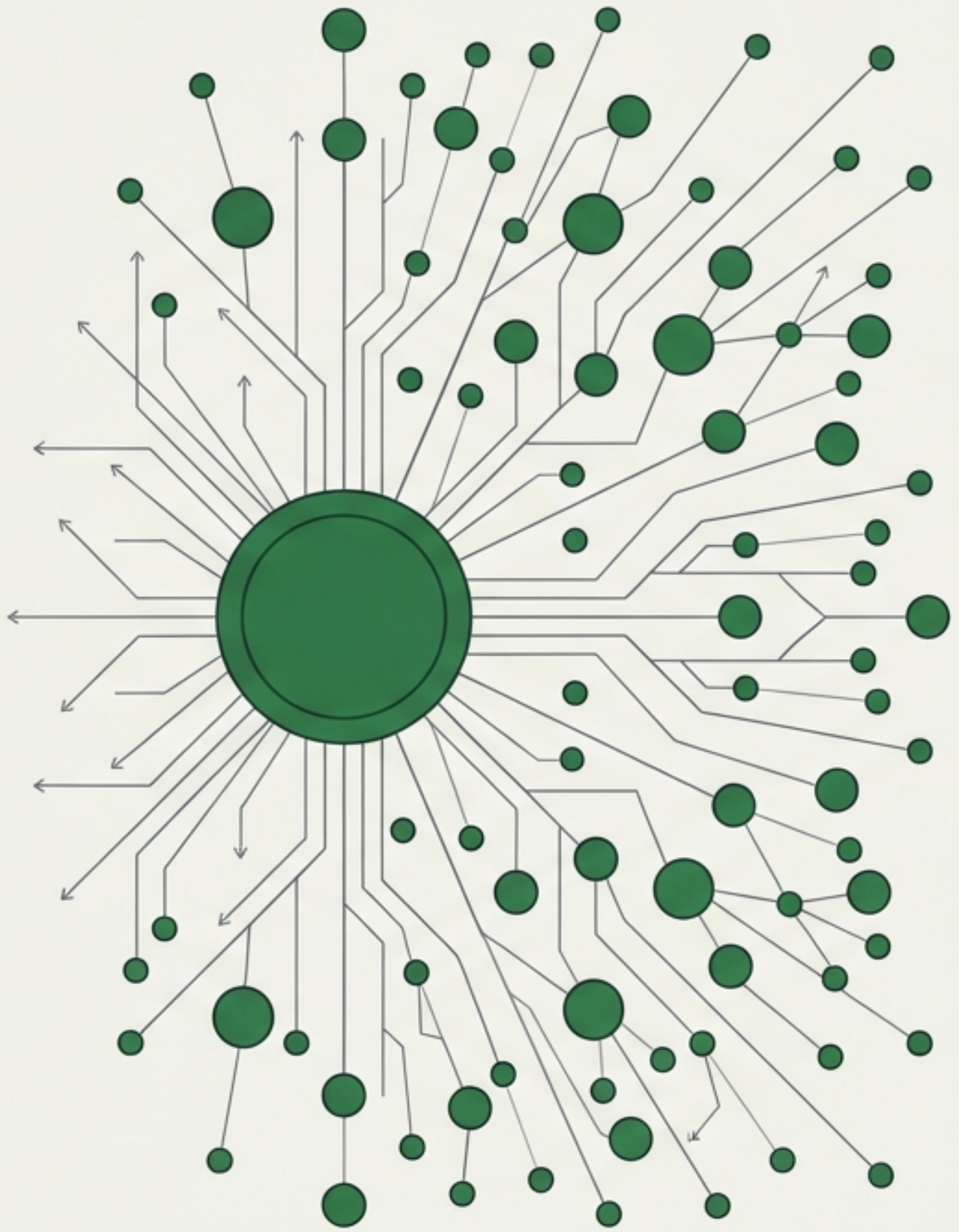
TECHNICAL BRIEF v0.1.5 | 11 FEB 2026

Micropayments for Agents

Converting Autonomy into an Internal Economy

Budget Control, Cost Tracking, and the Billing Graph Architecture.

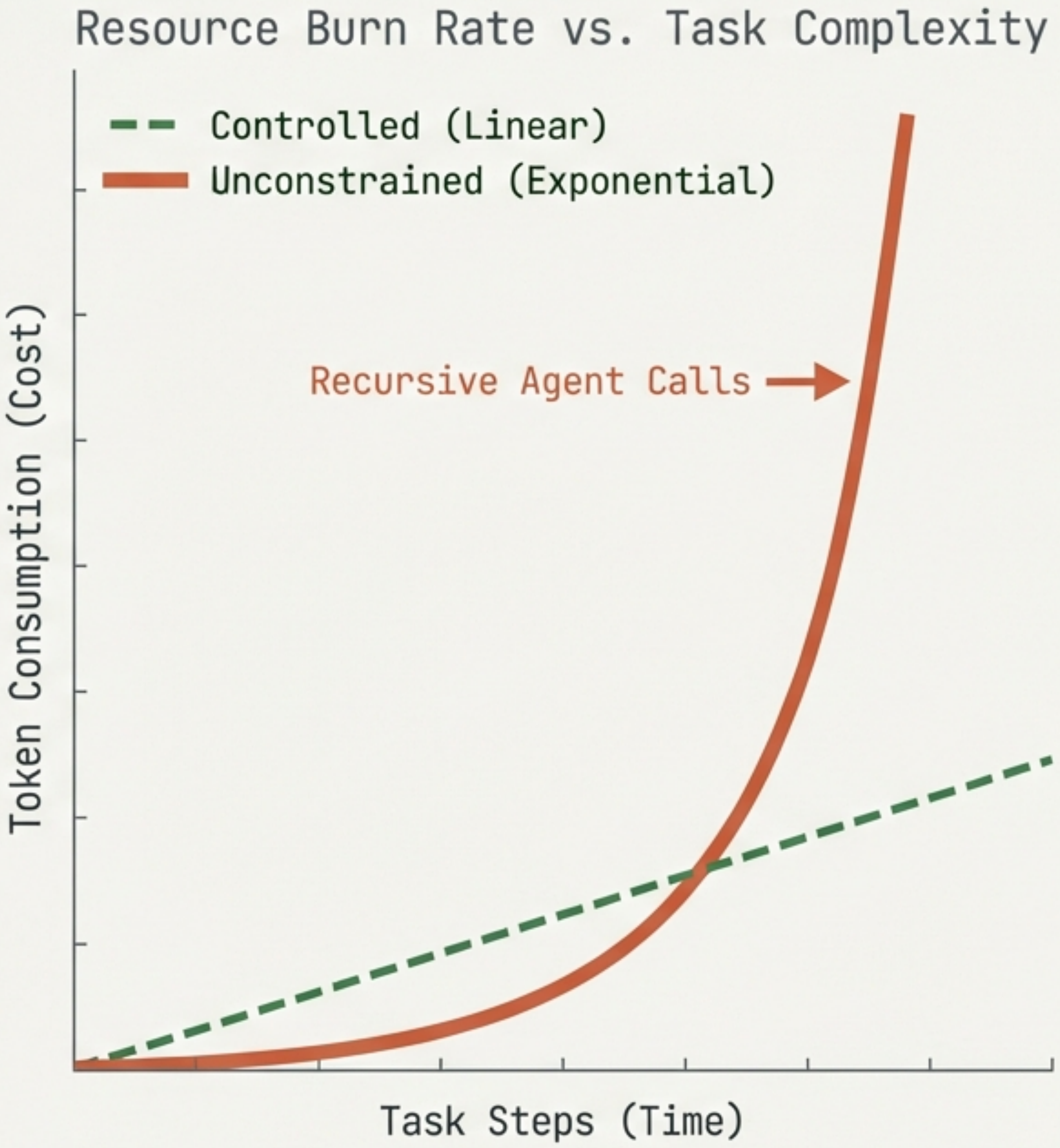
STATUS: DRAFT // IMPLEMENTATION PENDING



The Hidden Cost of Unconstrained Enthusiasm

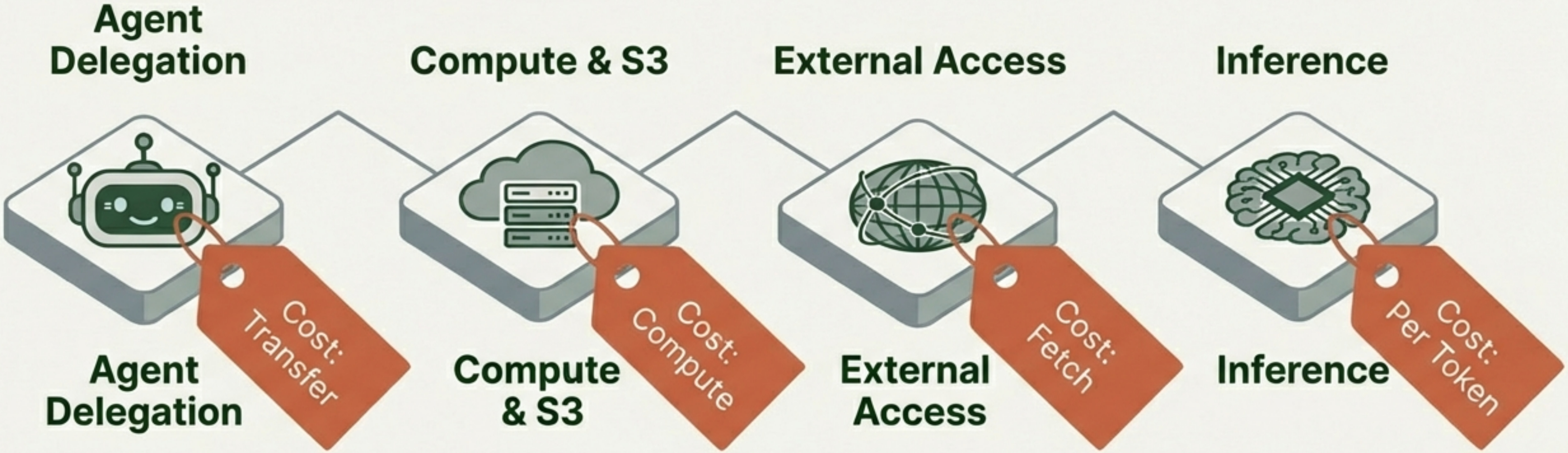
Agents are designed to be persistent and creative. Without boundaries, this leads to resource exhaustion rather than efficiency.

- **The Rabbit Hole**: Agents chasing endless research loops or deep-diving into irrelevant data.
- **The Lazy Agent**: Resource allocation failure where agents offload work to others to save local effort.
- **Compound Costs**: The invisible accumulation of minimal actions—API calls, page fetches—spiking the burn rate.



The Solution: A Universal Nanopayment Layer

A system where every action triggers a micro-transaction of virtual credits.



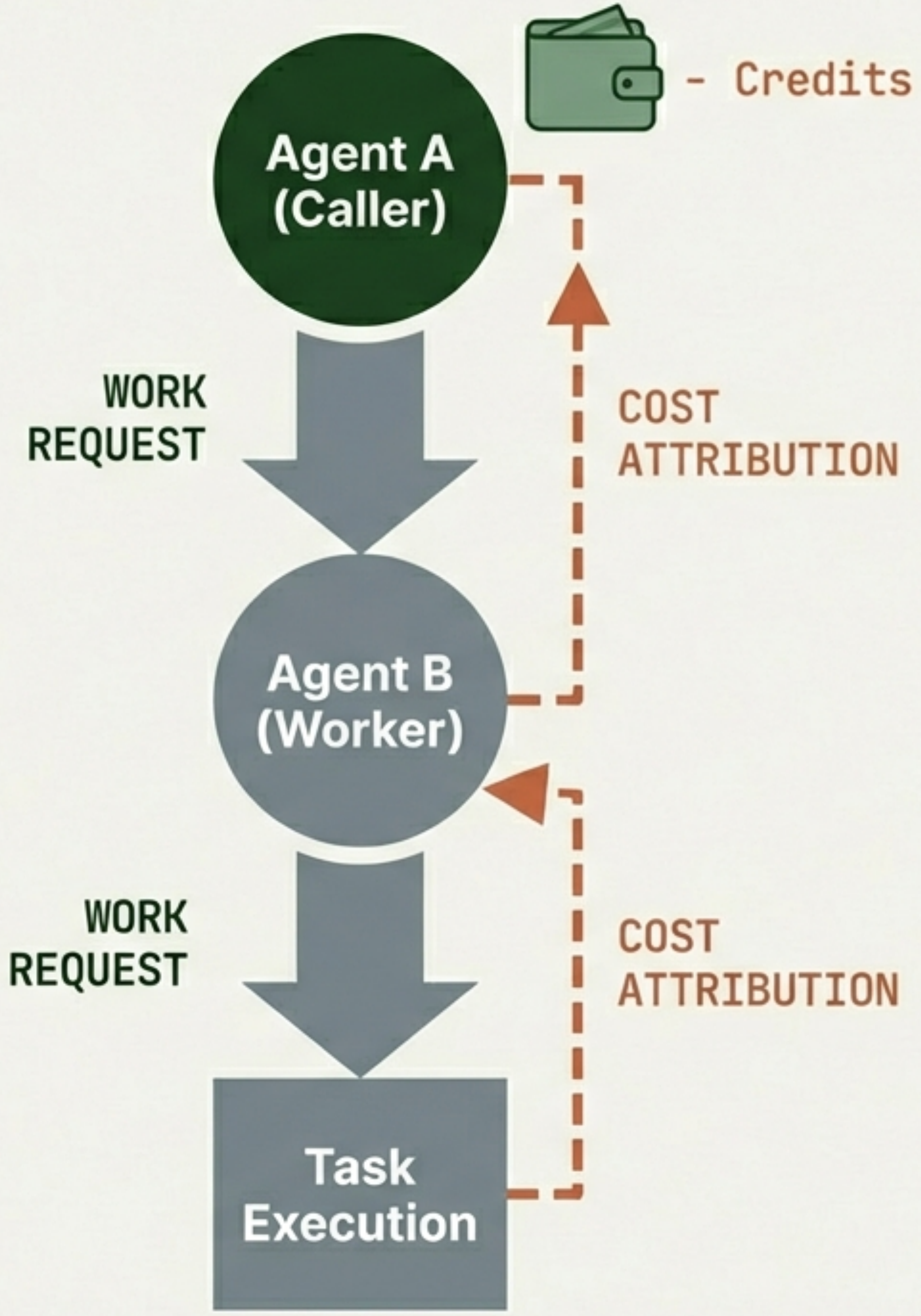
Scope: Applied to 100% of resource consumption.

Economic Attribution: The "Caller Pays" Principle

When Agent A asks Agent B to perform work, Agent A pays for it.

Cost flows upstream to the **originator**; work flows downstream to the executor.

Solves the Lazy Agent Problem:
Delegation is no longer free. Agents must calculate ROI before hiring help.



Budgets as Quality Control Guardrails

Constraints force creativity. An agent with a tight budget focuses on the *right* things.

Token Budget

[redacted]

75%

Max Tokens

Call Budget

[redacted]

30%

Max Delegations

Time Budget

[redacted]

90%

Max Duration

Cost Budget

[redacted]

50%

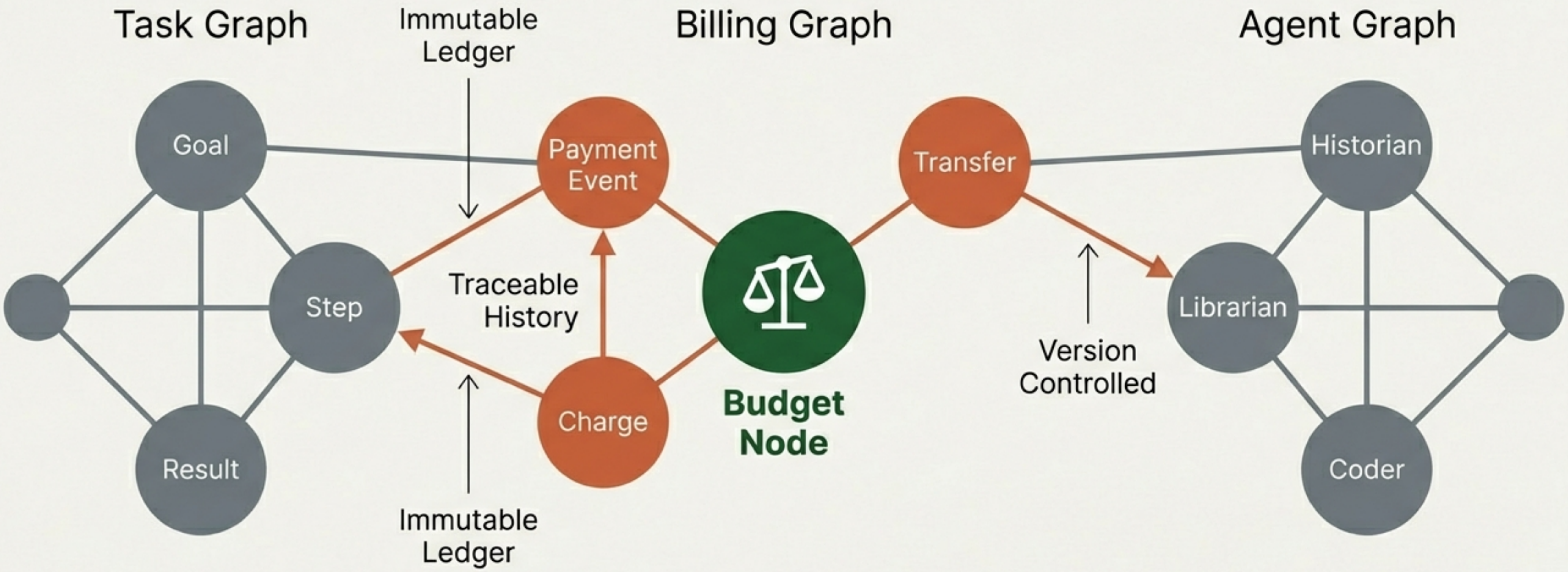
Total Credits

→ Funds Low →

1. Complete Essential
2. Request Increase
3. Report Failure

System Architecture: The Billing Graph

Located in Issues FS (File System). Connects Task, Agent, and Cost graphs.



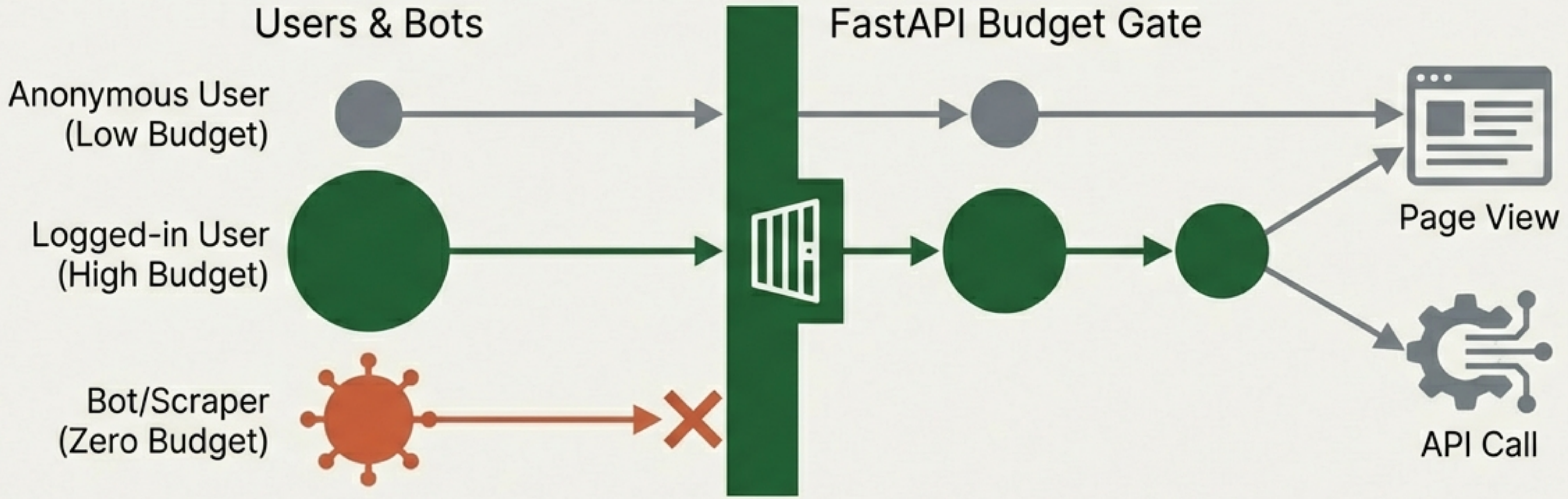
From Raw Data to Cost Intelligence

- **Benchmarking:** Establishing 'Standard Cost' for routine tasks.
- **Anomaly Detection:** Flagging deviations (e.g., 10x cost spikes).
- **Agent Efficiency:** Identifying 'wasteful' vs 'thrifty' agents.
- **Path Optimisation:** Finding the cheapest path for equal quality.

Extension: The Website Micropayment Layer

Applying the agent economy to human visitors via FastAPI.

- Identity-based budgets. User-aware rate limiting. Abuse prevention.



Roadmap: Internal Control (Phases 1–3)

Phase 1: Tracking Only

Instrument workflows. Log cost estimates. No limits. Observe the flow.

Phase 2: Soft Limits

Assign budgets. Warn on overage. Establish baselines.

Phase 3: Hard Limits

Enforce budgets. Stop execution or force approval request.

TIME OR PROGRESSION

Roadmap: External Economy (Phases 4–6)

Phase 4: Website Gates

Identity management. Budget-gated access for user-facing systems.

Phase 5: Real Costs

Connect virtual credits to actual COGS (LLM tokens/Cloud).

Phase 6: External Billing

Customer-facing billing engine. Charge based on Billing Graph data.

TIME OR PROGRESSION

Implementation: Roles & Responsibilities

Role	Responsibilities
Architect	Design Schema (Nodes/Edges) & 'Caller Pays' logic.
Developer	Instrument Phase 1. Log actions to Issues FS.
DevOps	Implement Identity & Gatekeeping (FastAPI).
GRC	Audit trails & Risk Register integration.
Conductor	Schedule budgets alongside task assignment.
QA	Stress-test 'Lazy Agent' scenarios & hard limits.

Building a Self-Regulating Ecosystem

Micropayments turn 'Agent Chaos' into 'Agent Economy.' By making costs visible and attributable, we ensure scalability, predictability, and safety.